Update 18/08/2026: StealthWriter can now remove the Claude watermark from AI-generated text

View Changelog

Humanizer

Humanizer

How to use

Paste your AI-generated text below and humanize it.

Level 1 2 3 4 5 6 7 8 9 10 Heavy rewrite with maximum variation

Model Ghost 5.2 Mini Ghost 5.2 Pro Legacy

generates a “affected region” visualization, which is “best effort” at this time. from the Vision Transformer attention weights of the classifier, is used as a FastAPI backend that is linked to a React frontend. additionally offers treatment/prevention advice, diagnosis The water-quality monitoring panel, history, and a water-quality improvement program. We report the The actual and actual measured performance and current limitation of system While we should be honest and talk about the RT-DETRv2-based de- The classical-CV localization stage was the one that was targeted by tector. This is the actual used, as this functionality is not present in bounding-box- annotated training data. Keywords—fish disease diagnosis, aquaculture, hybrid deep

372/1,000 words Clear Limited daily humanizations Limited daily scans resets at 5:30 AM (6h 5m)

Check for AI Humanize

Advanced Settings

Humanized Result

Humanize More Rehumanize Deep Scan On Compare Copy

Subsections come in two distinct data classes.There are two types of low data classes for subsections.
Although the specificity for both classes is 1.000, the precision, recall and F1 scores for \emph{Ulcer_Disease} and \emph{Tail_And_Fin_Rot} are all 0.000, which means that because the number of training images for both classes is small, the model is never able to predict either class on the test set, neither correctly nor as misclassifications. This is all the explanation for why the macro-average F1 (0.760, which gives all nine classes the same weight because of how it is calculated) is much lower than the very high weighted-average F1 (0.968, which means that these 2 classes have only 7 of 780 test images, or under 1%). Instead of trying to avoid reporting the macro metrics, or only reporting accuracy, we report the macro metrics because the class-imbalance problem is only apparent when the macro metrics are reported. This is a data-volume limitation, and not an architecture limitation—the same model architecture achieves F1 $>$ 0.95 on all classes with similar training data volume as the others.

We qualitatively examined the level of localization and the disease-attention overlay.The level of localization and the disease attention overlay were qualitatively assessed.
The fish-localization stage and the stage of visualizing the disease did not have any ground truth bounding boxes or any disease masks in the dataset, so both were evaluated qualitatively during development, not quantitatively, for instance by calculating IoU. The edge-span heuristic (Section~III-B) that is used in the localization stage was tested with an assortment of synthetic test cases: a fish silhouette on a plain background, a close-up filling the entire frame and a frame divided by strong internal shading and the edge-span heuristic correctly avoided cropping into the fish in all three. The fish-localization box is always good at catching the entire fish when applied to real photographs. Secondly, the disease-attention overlay is not as reliable in a qualitative sense; for real disease photos that were tested during development, the highlighted region often appears on a reasonable portion of the fish and sometimes on a part that doesn't correspond to the clearly diseased area. We believe this is due to: (1) the coarse attention grid of 14×14; (2) no attention-level supervision during training; and (3) the attention signal of the CLS token being a global-context signal, not a class-conditional saliency map as constructed. It is presented to the user as a best-effort visual aid (not a localization worthy of a diagnosis) and is considered a future work priority, not a solved problem, for improvement -- most plausibly through a small amount of lesion-level annotation.

We summarize the honest limitations of the current system: (1) Only 9 of the 11 target disease classes have any useable public data, and among these 9, 2 classes have too little data to be reliably learned, (2) the fish-localization stage is classical computer vision, not the target detector proposed by the authors, RT-DETRv2, because there is no public dataset for this task with bounding box annotations, (3) the disease-attention visualization is an off-the-shelf attention mechanism, not a validated lesion segmentation because there is no subject/farm

identifier in the source datasets, (4) the train/validation/test split is done at image level, not the subject/farm level, because there is no subject/farm identifier in the source datasets, and (5) the water-quality panel is simulated, not supported by physical sensors.

We tested our deployed hybrid EfficientNet-B4/ViT-B16 fish disease diagnosis system on a 9-class, 780-image held-out test set, achieving a test accuracy of 0.9718 and a macro F1 score of 0.76 which we argue is due to two data-starved classes, rather than the model. In addition to the classifier, the system includes a fish-presence gate, treatment guidance, a water-quality panel, a diagnosis history and full-stack deployment, as well as an attention-based disease-region visualization. We reported the difference between this system and the targeted 11-class, RT-DETRv2-detected design explicitly, not pretending a substitution classical-CV localization stage as a trained detector.

In the future, more images of class \emph{Ulcer_Disease} and \emph{Tail_And_Fin_Rot} could be collected or annotated, and the two classes completely missing from the database could be collected or annotated, which would lead to a more robust real RT-DETRv2 detector replacing the classical-CV localization stage, and finally, a small subset of images at the lesion level could be collected to properly train and test the disease-region visualization, without having to depend on unsupervised attention.

Green = high human impact, Red = high AI impact. Click sentences to swap alternatives. 732 words

Detection Score

Scan with **V1** V2 V1

ORIGINAL

Scanned with V2

100%

human

Looks Human

HUMANIZED

Scanned with V1



Looks Human

Looks Human